

Sandeep Satheesh

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EDUCATION

University of Illinois, Urbana-Champaign <i>Master of Computer Science</i>	Urbana-Champaign, Illinois 2026 – 2028
Indian Institute of Technology, Madras <i>Bachelor of Science (B.Sc) - Programming & Data Science; GPA: 3.6/4.0</i>	Chennai, India 2022 – 2026
Anna University <i>Bachelor of Engineering (B.E) - Electronics and Communication Engineering; GPA: 3.73/4.0</i>	Chennai, India 2021 – 2025

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, DBMS (PostgreSQL, MySQL, MongoDB), Neo4j, HTML
Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Pandas, PydanticAI, FastAPI, React, LangChain, LangSmith
Tools: Docker, Kubernetes, Git, Azure DevOps, AWS, GCP, Supabase, Redis, GitHub, TinyML

PROFESSIONAL EXPERIENCE

S2T AI AI-Powered Investigations <i>Software Engineer L1, prev. Software Engineer Intern</i>	Singapore Sept 2025 – Aug 2026
- Streamlined data ingestion by engineering APIs to ensure consistent data collection, and architected an Agentic AI (MCP Driven) workflow to automate backend integrations and NLP processing. - Scaled and maintained microservices via Kubernetes, managing robust DevOps/MLOps lifecycles and orchestrating end-to-end ML pipelines.	
Vessel Match (Start-up at Nirmaan IIT Madras) 	Chennai, India May 2024 – Aug 2025
<i>Founding Engineer - Head of Research & Workflows</i> - Worked on R&D of a production-ready multi-agentic workflow, combining RAG + agent orchestration to automate end-to-end maritime logistics and data processing. - Worked on event-driven microservices architecture using AWS (Lambda, EC2) + FastAPI for a production platform. - Scaled pipelines to process 200+ vessels/day and securing \$2K AWS credits.	
Indian Institute of Technology, Madras <i>Teaching Assistant - Business Data Management (MS2001)</i>	Chennai, India Jan 2024 – Sept 2025
Facilitated doubt-clearing sessions, curated datasets, and built Python + LLM evaluation scripts to automate submission checks, improving grading speed and data integrity.	
Paradox, IIT Madras <i>Steering Committee, Paradox' 2026 & Secretary, Paradox' 2025</i>	Chennai, India Jan 2023 – Present
Steering Committee for Paradox '26 driving financial audits and planning, and having led Paradox '25 as Secretary managing 14 departments and 50+ events, overseeing logistics and setup for 5,000+ attendees with a INR 16M budget.	

PROJECTS

GuruHeal  Presentation 	May 2025
AI-powered wellness platform with multilingual chat, prompt-engineered LLM pipelines, real-time retrieval, and personalized recommendations. Microservices with a GraphRAG knowledge base, agentic web search , tool-calling , and Generative UI. Next.js · PydanticAI · TypeScript · JavaScript · Figma · FastAPI · Supabase · Redis · Neo4j	
LLM Automation Agent 	Feb 2025
LLM-powered agent that takes plain-English task descriptions and runs them automatically. When the requirement is provided the agent chooses and runs the right tools—SQL, OCR, embeddings, web scraping, file ops—and returns results. FastAPI · OpenAI · LangChain · PydanticAI · LiteLLM · OpenRouter · Docker · GitHub Actions	
Crime Analysis and Prediction 	June 2024
Geospatial analysis portal for the Karnataka Police Hackathon . Utilized Time Series Prediction and behavioral analysis to identify crime hotspots and trends across the state. Django · PyTorch · React · Facebook Prophet · Pandas · PyTorch · Tableau · Time Series · Folium	

RESEARCH & PUBLICATIONS

Modelling Clinical and Radiographic Data for Early Detection and Prediction of Osteoarthritis 	[ Paper]
<i>Published: IEEE CSNT '25 (VIT Bhopal)</i> Late-fusion: EfficientNetB3 (X-rays) + XGBoost (clinical), weighted ensemble for early KOA detection and grade prediction.	
KOA Prediction & Grading via Late-Fusion + LLM/RAG Insights 	[ Folder]
<i>Indian Patent App. No. 202541071393 A (Filed: 28 Jul '25; Pub: 01 Aug '25)</i> Late-fusion Model (EfficientNetB3 + XGBoost) with LLM+RAG Pipeline for interpretability and patient-facing insights.	

HONORS & AWARDS

- Winner — Karnataka State Police Datathon, 2024:** Won among 30,000+ initial submissions; built a geospatial + time-series crime analysis/prediction system for Karnataka State Police organized with support from Microsoft.
- Technical Competitions:** Secured multiple Winner and Runner-up titles at various national-level technical fests for circuit design, hardware debugging and Software Design contests.